
EXP 4

Aim: Creation of account in Metamask, Transfer of Amount/Cryptocurrency from one account to another.

Theory:

1. What is MetaMask?

MetaMask is a software-based cryptocurrency wallet and browser extension (also available as a mobile app) that lets users interact with the Ethereum blockchain and other EVM-compatible networks (like Polygon, BSC, Avalanche, etc.). It acts as a bridge between a regular web browser and decentralized applications (dApps), without requiring users to run a full blockchain node themselves.

2. Key Functions of MetaMask

- **Wallet Management:** Stores public and private keys, allowing users to send, receive, and hold cryptocurrencies and tokens.
- **Account Creation:** Generates a new wallet using a 12-word (or 24-word) mnemonic seed phrase, from which all keys are mathematically derived.
- **Transaction Signing:** Signs transactions locally using the private key before broadcasting them to the network — the private key never leaves the user's device.
- **dApp Connectivity:** Injects a Web3 provider ([window.ethereum](https://window.ethereum.org/)) into the browser, letting websites request wallet connections and transaction approvals through a popup interface.
- **Network Switching:** Supports multiple blockchain networks and lets users add custom RPC endpoints (e.g., testnets, Layer-2 chains).

3. How MetaMask Works Internally

- **Key Generation:** Uses the BIP-39 standard to create a seed phrase, and BIP-32/BIP-44 to derive a hierarchy of key pairs from that seed.
- **Public/Private Key Pair:** The private key is used to sign transactions; the public key (and the derived address) is what others use to send funds or verify signatures.
- **Local Storage:** Keys are encrypted and stored locally in the browser's storage, protected by a user-set password.
- **Non-Custodial Nature:** MetaMask does not store user funds on any server — the user has full control (and full responsibility) over their private keys and seed phrase.

4. Role in Blockchain Interaction

- Acts as a signer for transactions instead of the dApp itself, keeping private keys isolated from websites.
- Communicates with the blockchain via RPC (Remote Procedure Call) providers like Infura or Alchemy, or a self-hosted node.

- Enables smart contract interaction: users can approve token transfers, mint NFTs, vote in DAOs, or use DeFi protocols directly from the browser.

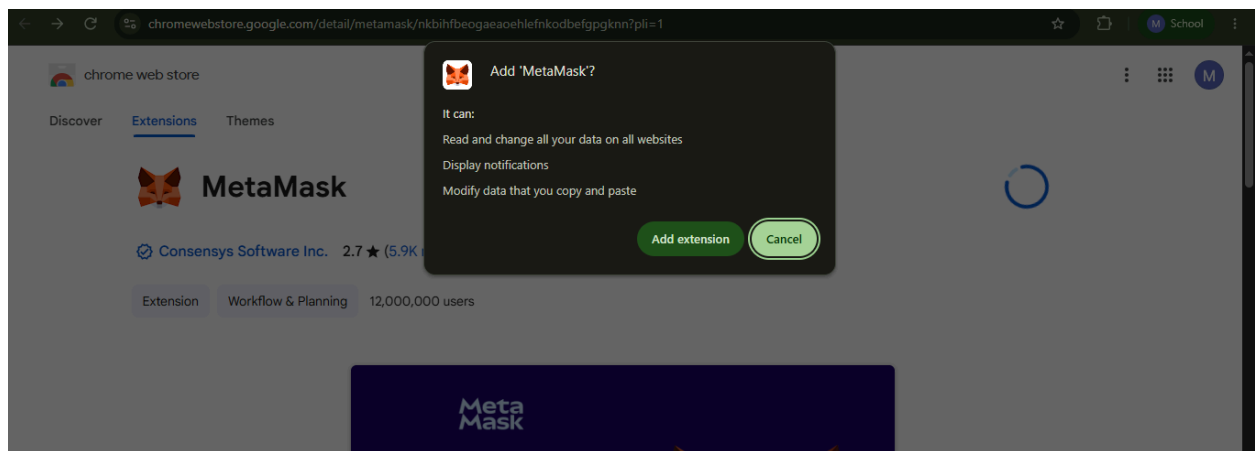
5. Security Considerations

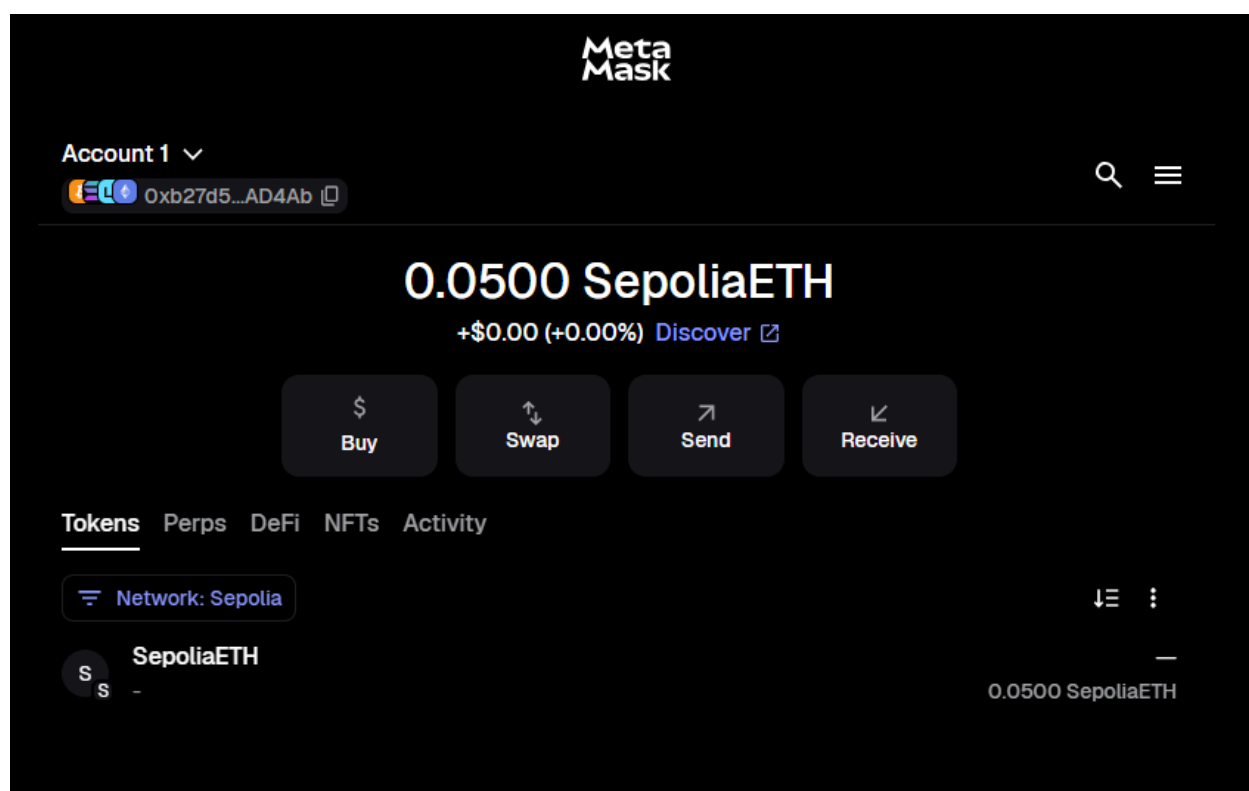
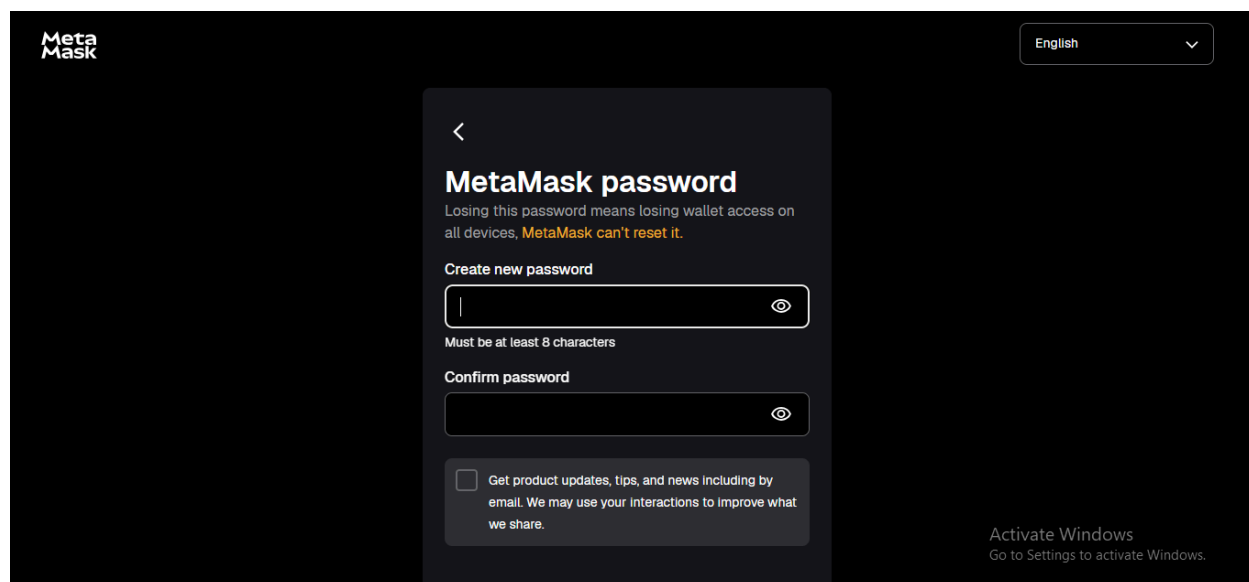
- The seed phrase is the single point of recovery — anyone with it can access the wallet, so it must be stored offline and never shared.
- MetaMask includes phishing detection, warning users about known malicious sites.
- Hardware wallet integration (e.g., Ledger, Trezor) is supported for extra security, keeping private keys on a separate physical device.

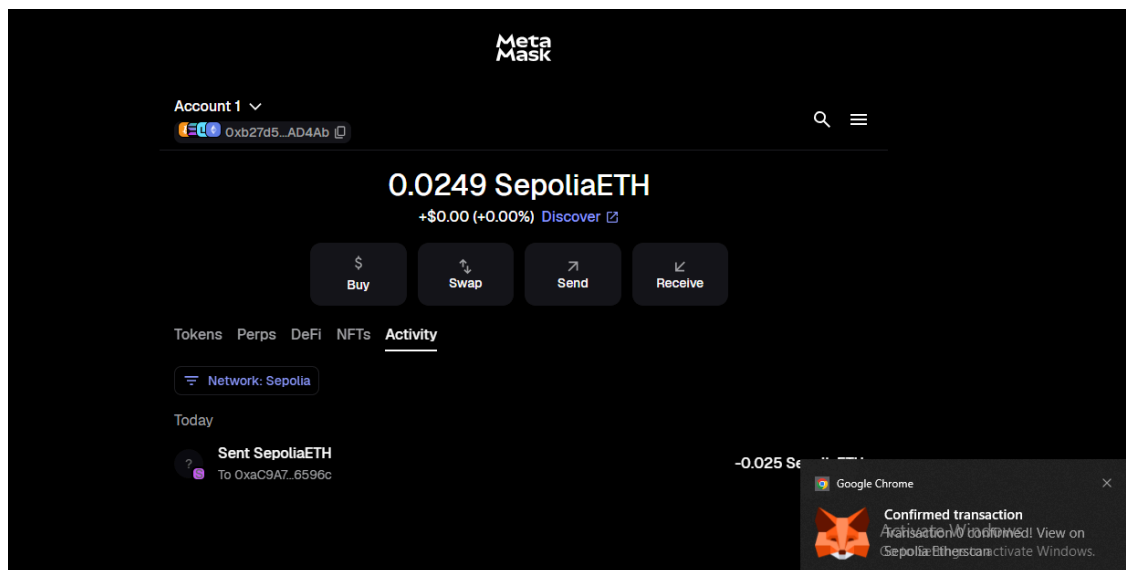
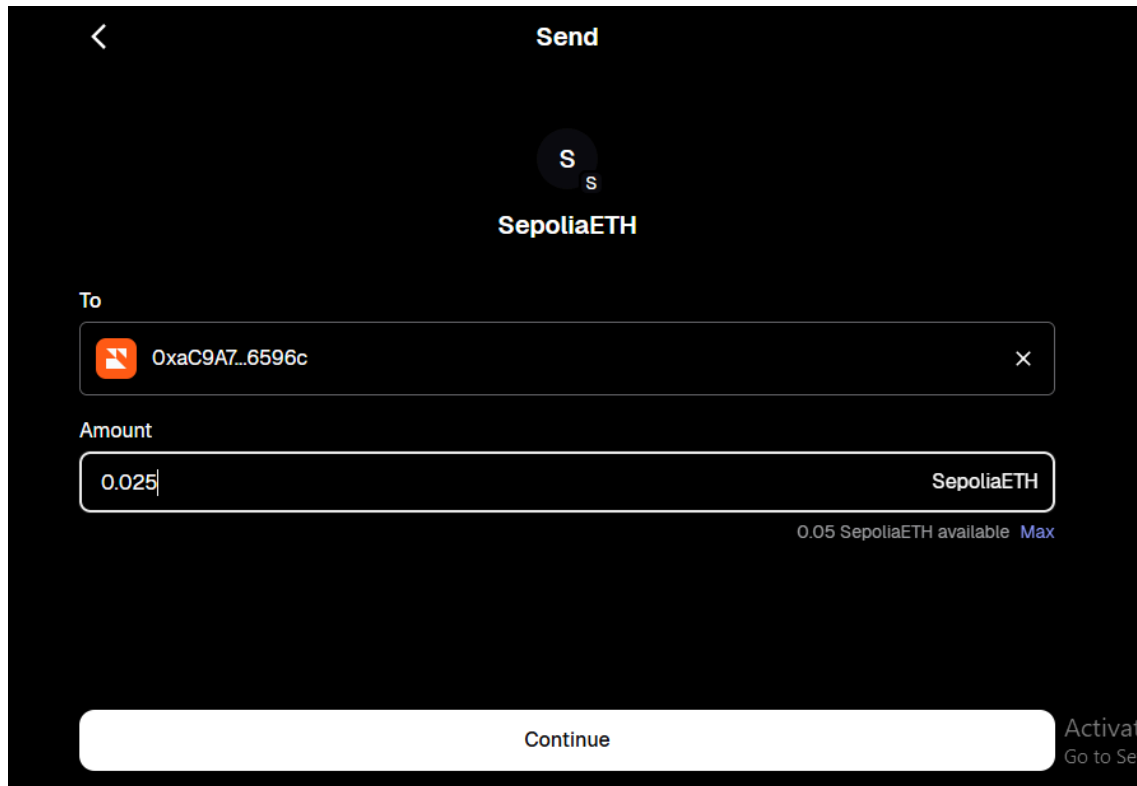
Steps:

1. MetaMask Setup & Account Creation
 - Installed the MetaMask extension and selected Create a New Wallet.
 - Set a local access password and backed up the 12-word Secret Recovery Phrase offline.
 - Verified the seed phrase to finalize wallet generation.
2. Connecting to Sepolia & Funding Wallet
 - Enabled Show Test Networks in MetaMask settings and selected Sepolia.
 - Copied the generated public account address (\$0x...\$).
 - Navigated to the Google Cloud Web3 Sepolia Faucet, pasted the address, and requested testnet SepoliaETH.
3. Executing the Peer-to-Peer Transfer
 - Selected Send in MetaMask and pasted the recipient's wallet address.
 - Specified the amount of SepoliaETH to transfer.
 - Reviewed network gas fee estimations and clicked Confirm to sign and broadcast the transaction.
 - Tracked transaction confirmation status via the block explorer (Sepolia Etherscan).

Output:







Conclusion

This experiment confirms the operational flow of decentralized value transfer without reliance on a central server or clearinghouse. Utilizing the Sepolia testnet via Google's Web3 faucet provides a zero-risk sandbox environment to observe cryptographic signing, gas fee dynamics, and block confirmation latency in real time.